

Lab-Assignment 6.4

NAME: P.RANADHEER REDDY

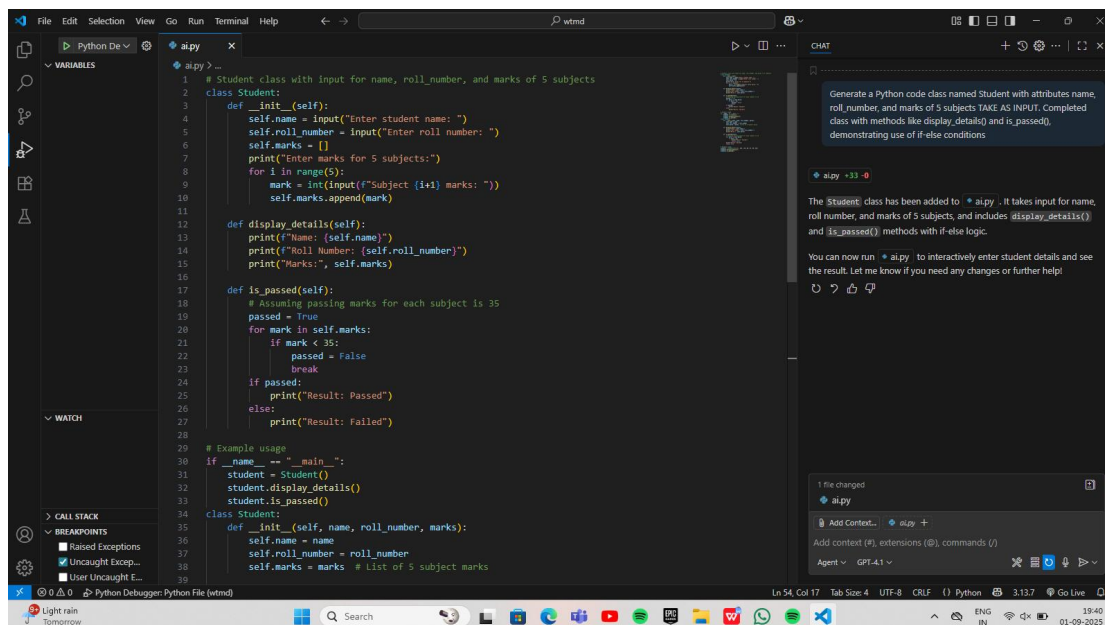
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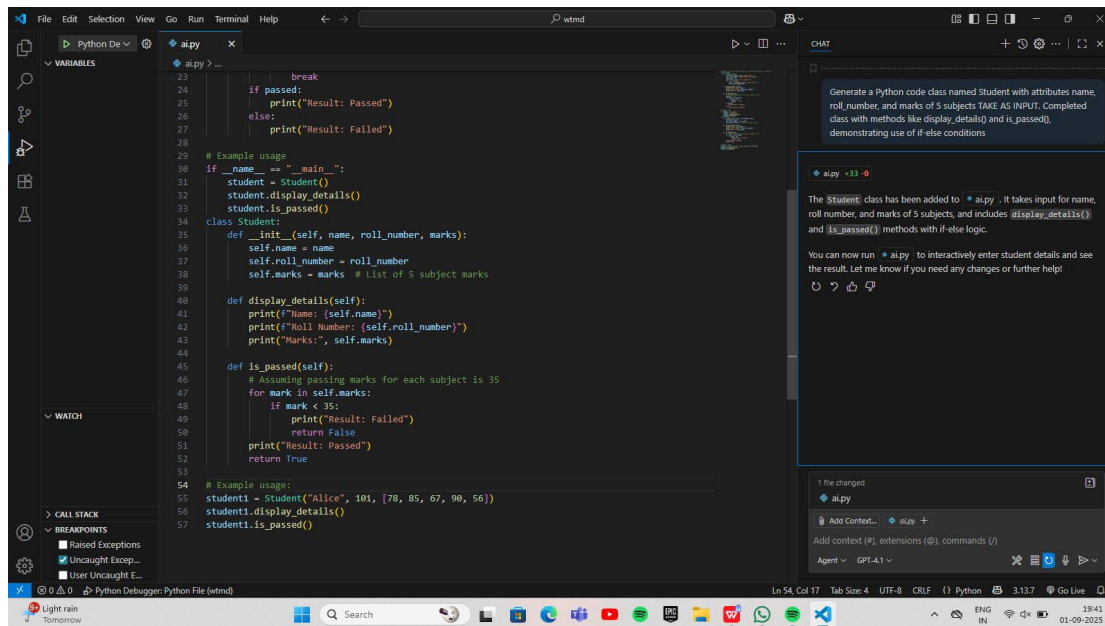
Task-1:

PROMPT: Generate a Python code class named Student with attributes name, roll_number, and marks of 5 subjects . Completed class with methods like display_details() and is_passed(), demonstrating use of if-else conditions.

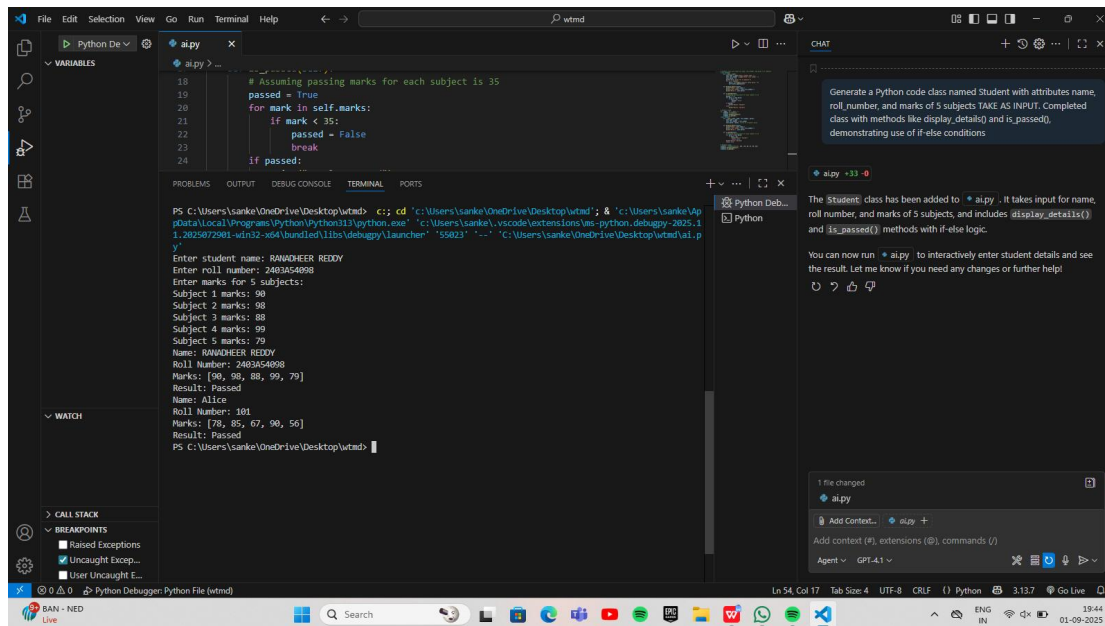
CODE:



```
1 # Student class with input for name, roll_number, and marks of 5 subjects
2 class Student:
3     def __init__(self):
4         self.name = input("Enter student name: ")
5         self.roll_number = input("Enter roll number: ")
6         self.marks = []
7         print("Enter marks for 5 subjects:")
8         for i in range(5):
9             mark = int(input(f"Subject {i+1} marks: "))
10            self.marks.append(mark)
11
12    def display_details(self):
13        print(f"Name: {self.name}")
14        print(f"Roll Number: {self.roll_number}")
15        print("Marks:", self.marks)
16
17    def is_passed(self):
18        # Assuming passing marks for each subject is 35
19        passed = True
20        for mark in self.marks:
21            if mark < 35:
22                passed = False
23                break
24        if passed:
25            print("Result: Passed")
26        else:
27            print("Result: Failed")
28
29    # Example usage
30    if __name__ == "__main__":
31        student = Student()
32        student.display_details()
33        student.is_passed()
34
35    class Student:
36        def __init__(self, name, roll_number, marks):
37            self.name = name
38            self.roll_number = roll_number
39            self.marks = marks # List of 5 subject marks
```



OUTPUT:

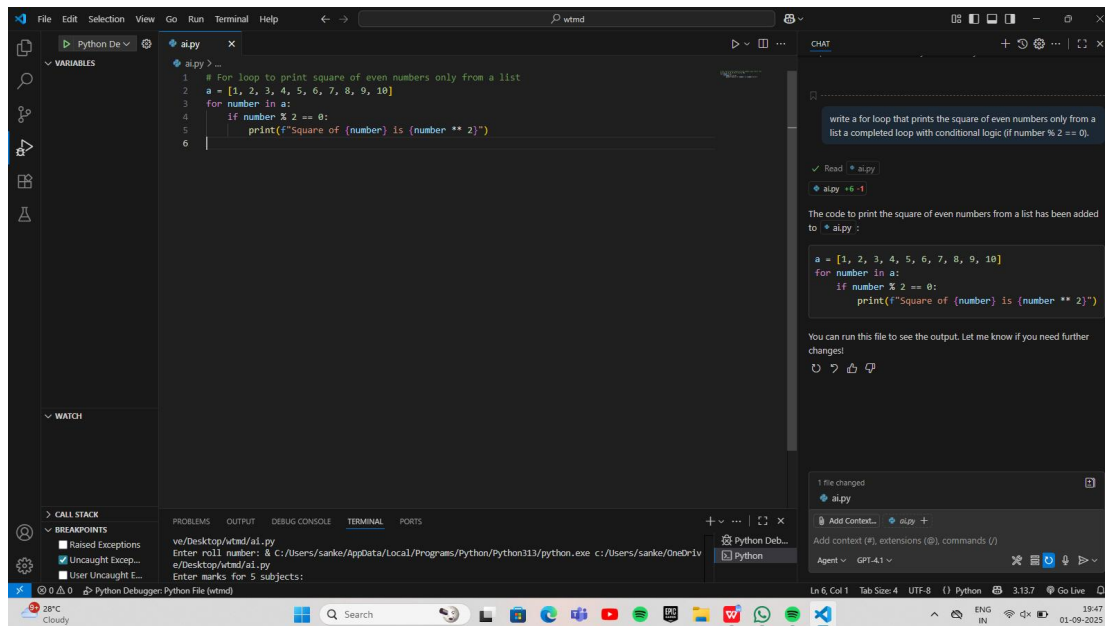


Task-2:

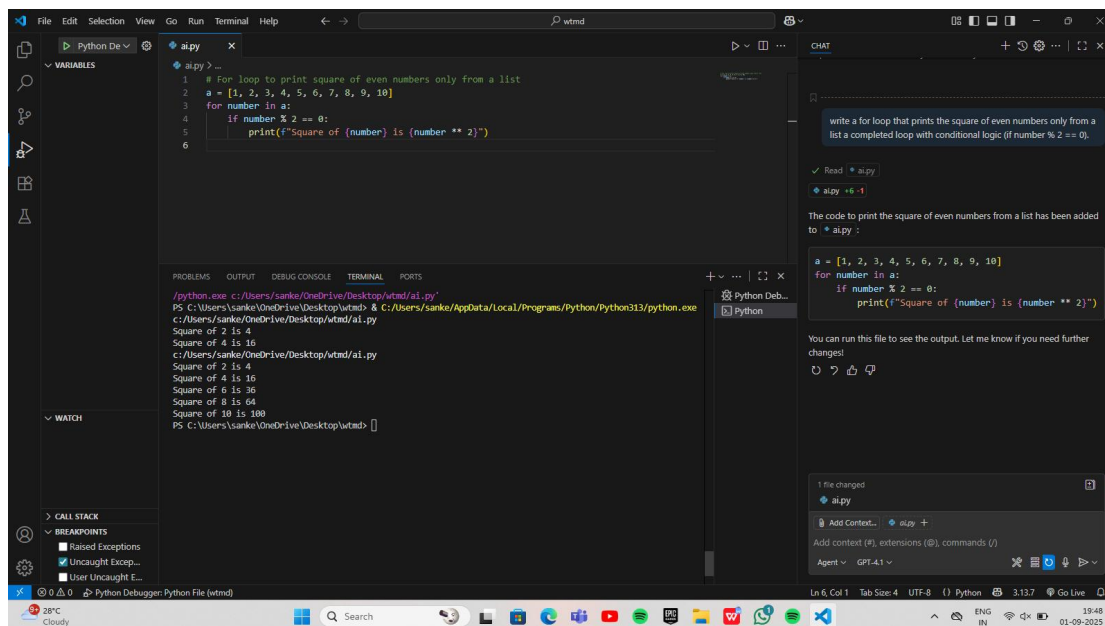
PROMPT:-

write a for loop that prints the square of even numbers only from a list a completed loop with conditional logic (if number % 2 == 0).

CODE:



OUTPUT:



Task-3:

PROMPT:

I need to generate a python code a Functional class with complete method definitions using if conditions and self attributes. with class called BankAccount with attributes account_holder and balance using methods deposit(), withdraw(), and check for insufficient balance

CODE :-

```
class BankAccount:
    def __init__(self):
        self.account_holder = input("Enter account holder name: ")
        self.balance = float(input("Enter initial balance: "))

    def deposit(self):
        amount = float(input("Enter amount to deposit: "))
        if amount > 0:
            self.balance += amount
            print(f"Deposited {amount}. New balance: {self.balance}")
        else:
            print("Deposit amount must be positive.")

    def withdraw(self):
        amount = float(input("Enter amount to withdraw: "))
        if amount > self.balance:
            print("Insufficient balance.")
        elif amount <= 0:
            print("Withdrawal amount must be positive.")
        else:
            self.balance -= amount
            print(f"Withdrew {amount}. New balance: {self.balance}")

    def display_details(self):
        print(f"Account Holder: {self.account_holder}")
        print(f"Balance: {self.balance}")

# Example usage
if __name__ == "__main__":
    account = BankAccount()
    account.display_details()
    account.deposit()
    account.withdraw()
    account.display_details()
```

Balance: 1200.0
PS C:\Users\sanke\OneDrive\Desktop\wtmd>

CHAT

I need to generate a python code a Functional class with complete method definitions using if conditions and self attributes with class called BankAccount with attributes account_holder and balance using methods deposit(), withdraw(), and check for insufficient balance TAKE INPUT

ai.py +36 -1

The BankAccount class in ai.py now takes input for the account holder and initial balance, and its deposit() and withdraw() methods also take input interactively. It checks for insufficient balance using if conditions and self attributes.

Run the file to interact with the class. Let me know if you need further changes!

1 file changed
ai.py
Add context (#), extensions (@), commands (/)

Agent GPT-4.1

OUTPUT:

```
PS C:\Users\sanke\OneDrive\Desktop\wtmd> & C:/Users/sanke/AppData/Local/Programs/Python/Python313/python.exe c:/Users/sanke/OneDrive/Desktop/wtmd/ai.py
Enter account holder name: RANADHEER REDDY
Enter initial balance: 1000
Enter amount to deposit: 500
Deposited 500.0. New balance: 1500.0
Enter amount to withdraw: 300
Withdrew 300.0. New balance: 1200.0
Account Holder: RANADHEER REDDY
Balance: 1200.0
PS C:\Users\sanke\OneDrive\Desktop\wtmd>
```

CHAT

I need to generate a python code a Functional class with complete method definitions using if conditions and self attributes with class called BankAccount with attributes account_holder and balance using methods deposit(), withdraw(), and check for insufficient balance TAKE INPUT

ai.py +36 -1

The BankAccount class in ai.py now takes input for the account holder and initial balance, and its deposit() and withdraw() methods also take input interactively. It checks for insufficient balance using if conditions and self attributes.

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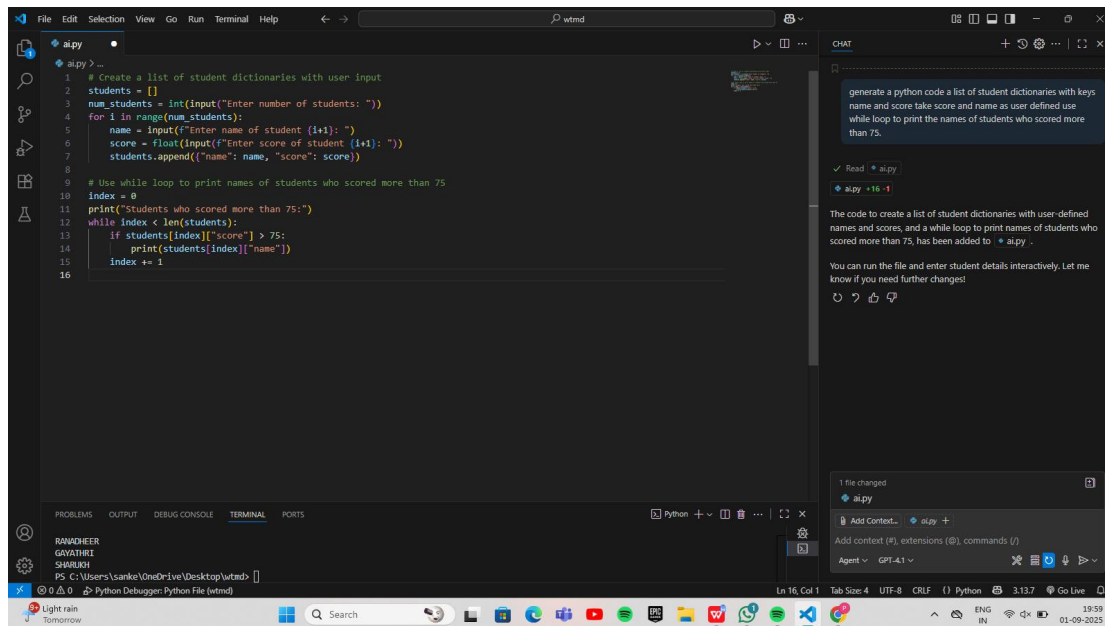
Agent GPT-4.1

Task4:

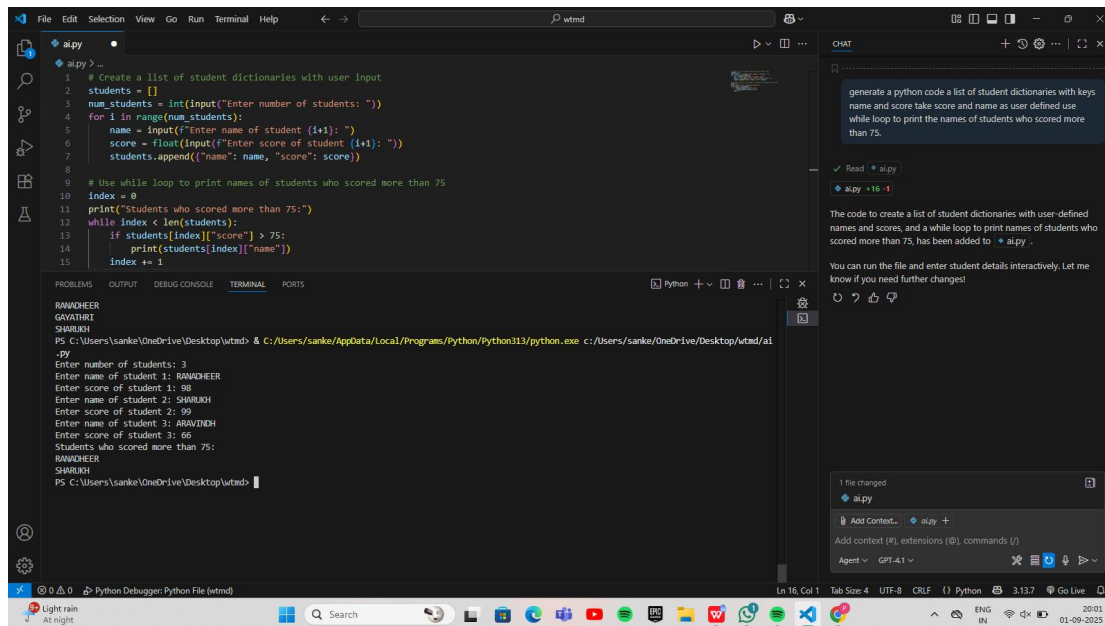
PROMPT:

generate a python code a list of student dictionaries with keys name and score take score and name as user defined use while loop to print the names of students who scored more than 75.

Code:



Output:



Task5:

Prompt:

Generate a python code with class ShoppingCart with an empty items list .Add methods to add_item, remove_item, and use a loop and if-else statements to calculate the total bill using conditional discounts

Code:


```
1 # ShoppingCart class with methods and conditional discount logic
2 class ShoppingCart:
3     def __init__(self):
4         self.items = [] # Each item is a dict: {"name": str, "price": float}
5
6     def add_item(self, name, price):
7         self.items.append({"name": name, "price": price})
8         print(f"Added {name} for {price}")
9
10    def remove_item(self, name):
11        for item in self.items:
12            if item["name"] == name:
13                self.items.remove(item)
14                print(f"Removed {name}")
15                return
16        print(f"Item {name} not found in cart.")
17
18    def calculate_total(self):
19        total = 0
20        for item in self.items:
21            total += item["price"]
22        # Conditional discount: 10% off if total > 500, else 5% off if total > 200
23        if total > 500:
24            discount = total * 0.10
25            print("Applied 10% discount.")
26        elif total > 200:
27            discount = total * 0.05
28            print("Applied 5% discount.")
29        else:
30            discount = 0
31            print("No discount applied.")
32        final_total = total - discount
33        print(f"Total bill after discount: {final_total}")
34        return final_total
35
```

Terminal Output:

```
Total bill after discount: 55000.0
PS C:\Users\sanku\OneDrive\Desktop\wtmd>
```

```
1 # ShoppingCart class with methods and conditional discount logic
2 class ShoppingCart:
3     def __init__(self):
4         self.items = [] # Each item is a dict: {"name": str, "price": float}
5
6     def add_item(self, name, price):
7         self.items.append({"name": name, "price": price})
8         print(f"Added {name} for {price}")
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10    def remove_item(self, name):
11        for item in self.items:
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15                return
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21            total += item["price"]
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25            print("Applied 10% discount.")
26        elif total > 200:
27            discount = total * 0.05
28            print("Applied 5% discount.")
29        else:
30            discount = 0
31            print("No discount applied.")
32        final_total = total - discount
33        print(f"Total bill after discount: {final_total}")
34        return final_total
35
36 # Example usage
37 if __name__ == "__main__":
38     cart = ShoppingCart()
39     cart.add_item("LAPTOP", 60000)
40     cart.add_item("MOUSE", 400)
41     cart.add_item("KEYBOARD", 1200)
42     cart.remove_item("MOUSE")
43     cart.calculate_total()
44
```

Terminal Output:

```
Total bill after discount: 55000.0
PS C:\Users\sanku\OneDrive\Desktop\wtmd>
```

Output:

